

- 6 A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected in series with a variable resistor and a uniform resistance wire XY, as shown in Fig. 6.1.

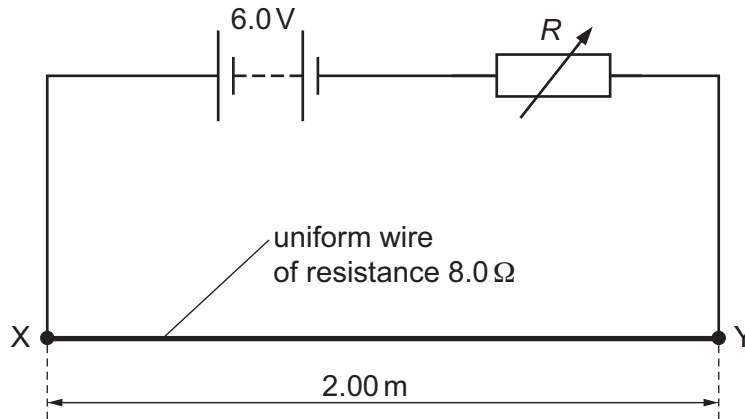


Fig. 6.1

Wire XY has length 2.00 m and resistance $8.0\ \Omega$. The resistance R of the variable resistor is adjusted so that the potential difference across wire XY is 2.4 V .

- (a) Determine R .

$R = \dots\dots\dots \Omega$ [2]

- (b) Explain why the potential difference V between any two points on wire XY is proportional to the distance L between those points.

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 [2]

